

PROJECT

5.8

ACCESS S3 FROM A VPC

NETWORKING SERIES

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Objective

This project focuses on setting up a VPC and launching an EC2 instance inside it. The goal is to create a secure, isolated environment for compute resources.

Next, we give the EC2 instance access to AWS services using access keys. Finally, we enable the instance to connect and interact with Amazon S3 for storage tasks.

Services

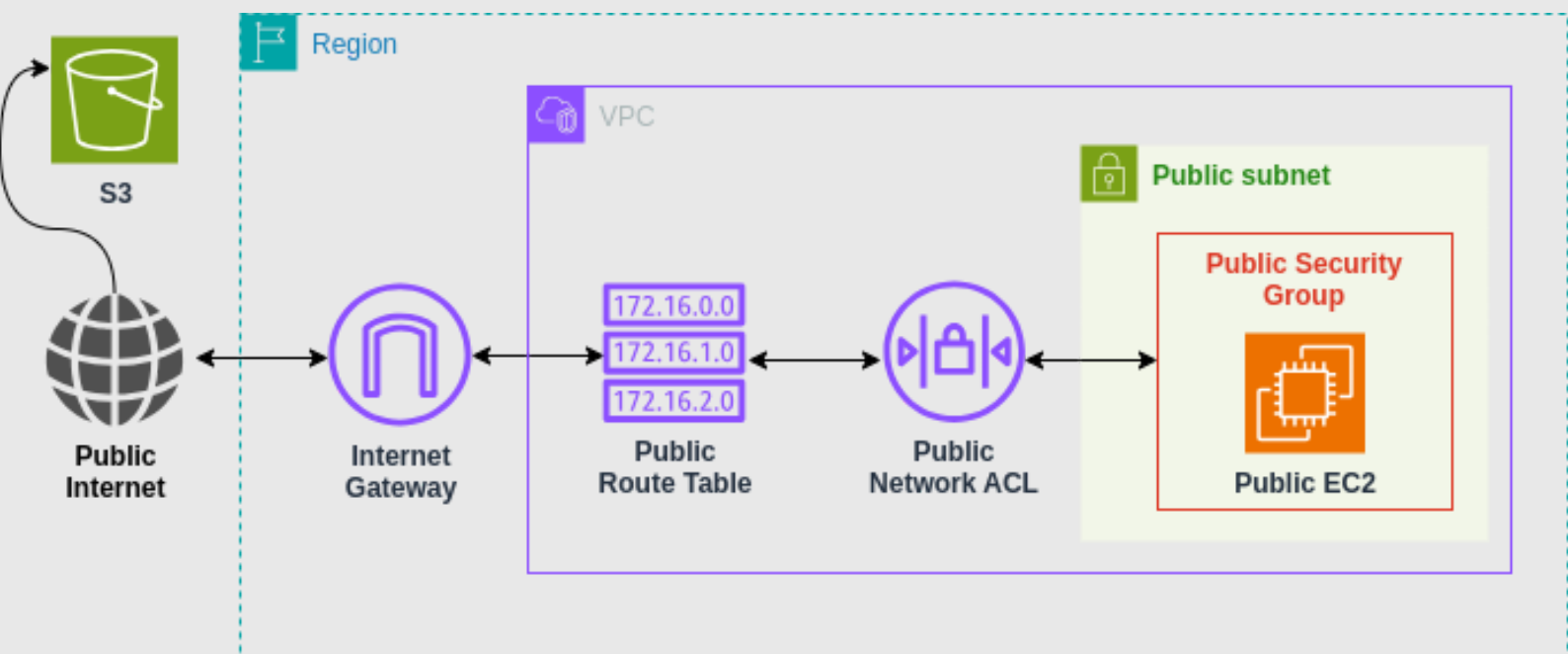
Amazon VPC - Creates an isolated cloud network where your EC2 instance runs securely.

Amazon EC2 - Launches a virtual server to perform compute tasks inside your VPC.

IAM (Access Keys) - Provides programmatic access so your EC2 can interact with AWS services.

Amazon S3 - Stores and retrieves files that your EC2 instance can access and manage.

Architecture Diagram



The architecture includes a VPC with one public subnet hosting an EC2 instance. The instance uses IAM access keys to securely connect to Amazon S3. This setup allows the EC2 to upload, download, or manage files stored in S3 from within a controlled network.

Steps to Implement

1. Create a VPC with Public Subnet:

Start by creating a VPC with a public subnet and attach an internet gateway to it. Then, update the route table so the subnet can access the internet through the gateway. This setup creates a secure network environment where your EC2 instance will run.

2. Launch an EC2 Instance:

Next, launch an EC2 instance within the public subnet, making sure it has a public IP address assigned. Select a key pair to securely connect to the instance via SSH from your local machine. This instance acts as your virtual server inside the cloud.

3. Generate IAM Access Keys:

Create an IAM user or role with the necessary S3 permissions and generate access keys for it. These keys provide your EC2 instance with the authority to access and interact with your S3 buckets. Keep these credentials secure as they control access to your AWS resources.

4. Connect EC2 to S3 Using AWS CLI:

Finally, connect to your EC2 instance using SSH and configure the AWS CLI using the IAM access keys. Use the CLI commands to list, upload, or download files from your S3 bucket directly through the EC2 instance. This enables seamless interaction between your compute and storage services.

Story Time

VPC (Your Cloud City):

Imagine building your very own cloud city with strong, protective walls called a VPC. This city keeps your resources safe and organized inside a private network. Your cloud computer (EC2) gets to live here, securely connected to the internet.

EC2 Instance (Your Cloud Computer):

Next, invite your cloud computer, an EC2 instance, into the city. It gets its own public IP address like a home address so you can visit and control it remotely. You use a special key pair as a secret handshake to safely enter and manage.

IAM Access Keys (Magic Access Pass):

To let your cloud computer interact with other cloud services, you give it a magic access pass IAM access keys. These keys grant it permission to open secret doors without risking your main passwords. This way, your EC2 can safely talk to the magical storage vault.

Amazon S3 (The Magical Storage Vault):

The magical storage vault, Amazon S3, keeps all your files safe and ready to use anytime. Your cloud computer can use its magic pass to fetch or store data here, acting like a trusted courier delivering packages inside the cloud city. This setup makes your cloud life secure and smooth.

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THANK YOU!

